Day 2: SQL Coding Challenge – Online Bookstore 📚

Question 1: CREATE TABLE with PRIMARY & FOREIGN KEY

Scenario:
You are creating a database for an online bookstore.

Task:

- Create a table Books with columns:
  - BookID → **INTEGER, PRIMARY KEY**
  - Title → **VARCHAR(100), NOT NULL**
  - Author → **VARCHAR(50), NOT NULL**
  - ISBN → **VARCHAR(20), UNIQUE**
  - Price → **DECIMAL(8,2), CHECK(Price > 0)**
- Create a table Orders with columns:
  - OrderID → **INTEGER, PRIMARY KEY**
  - BookID → **INTEGER, FOREIGN KEY REFERENCES Books(BookID)**
  - OrderDate → **DATE, NOT NULL**
  - Quantity → **INTEGER, CHECK(Quantity > 0)**

Expected Output:
Tables are created successfully with all constraints applied.

Question 2: ALTER TABLE – Add Default Constraint

Scenario:
The bookstore wants to make sure **ISBN is Default** for every book.

Task:

- Alter the Books table to **add a Default constraint** to the ISBN column.

Expected Output:
The ISBN column enforces uniqueness.

Question 3: INSERT, RETRIEVE & UPDATE with Constraints

Scenario:
You want to add sample book data and update certain records.

Task:

- Insert at least **5 records** into the Books table, respecting all constraints.
- Retrieve all records to verify entries.
- Update the Price or Quantity for a specific record **while maintaining the CHECK constraints**.

Expected Output:
All entries and updates comply with constraints and are displayed correctly.

Question 4: DELETE vs TRUNCATE

Scenario:
The bookstore wants to manage orders by removing some rows or clearing all data.

Task:

- Use **DELETE** with a **WHERE clause** to remove specific rows from Orders table.
- Use **TRUNCATE** to remove all rows while keeping table structure intact.

Expected Output:

- DELETE removes selected rows.
- TRUNCATE clears all rows quickly but keeps the table structure.